

CLOUD COMPUTING

What is Cloud? **Cloud Computing**

The term **Cloud** refers to software and services that run on the internet, instead of locally on the user computer.

Applications such as **e-mail, web conferencing, customer relationship management, DropBox** all run in cloud.

What is Cloud Computing?

Cloud Computing refers to **manipulating, configuring, and accessing** the applications online. offers online data storage, infrastructure and application.

WHY CLOUD COMPUTING

- **Traditional business applications have always been complex and costly.** They require a wide range of hardware and software, making setup and maintenance challenging.
- A dedicated team of specialists is often needed to install, configure, test, operate, secure, and regularly update these applications.
- When this process is repeated across dozens or even hundreds of applications, it becomes clear why even large organizations with advanced IT departments struggle to keep up with their needs.
- For small and medium-sized businesses, managing such applications is often unrealistic and nearly impossible.

With cloud computing, the burden of managing hardware and software disappears.

The shared infrastructure functions like a utility—you pay only for what you use, upgrades happen automatically, and scaling up or down is simple.

Cloud applications are faster and more affordable to deploy, often running within days or weeks.

All you need is a browser: log in, customize, and start using the app instantly.

BENEFITS

- Cost**

Eliminates the capital expense of purchasing hardware/software and reduces installation costs.

- Speed**

Large amounts of computing resources can be provisioned within minutes.

- Global Scale**

Provides the ability to scale elastically—delivering the right amount of IT resources when needed, from the right geographic locations.

- Productivity**

Removes the need for hardware setup, software patching, and other time-consuming IT management tasks, allowing businesses to focus on core goals.

BENEFITS

- Performance**

Runs on a worldwide network of secure data centers, regularly upgraded with the latest fast and efficient computing hardware.

- Reliability**

Simplifies data backup, disaster recovery, and business continuity by replicating data across multiple redundant sites in the provider's network.

- Security**

Cloud providers offer robust policies, technologies, and controls that enhance security, protecting applications, infrastructure, and data from threats.

S.No.	CLOUD COMPUTING	DISTRIBUTED COMPUTING
01.	Cloud computing refers to providing on demand IT resources/services like server, storage, database, networking, analytics, software etc. over internet.	Distributed computing refers to solve a problem over distributed autonomous computers and they communicate between them over a network.
02	In simple cloud computing can be said as a computing technique that delivers hosted services over the internet to its users/customers.	In simple distributed computing can be said as a computing technique which allows to multiple computers to communicate and work to solve a single problem.

03.	There are many benefits of cloud computing like cost effective, elasticity and reliable, economies of Scale, access to the global market etc.	There are many benefits of distributed computing like flexibility, reliability, improved performance etc.
04.	Cloud computing provides services such as hardware, software, networking resources through internet.	Distributed computing helps to achieve computational tasks more faster than using a single computer as it takes a lot of time.
05.	The goal of cloud computing is to provide on demand computing services over internet on pay per use model.	The goal of distributed computing is to distribute a single task among multiple computers and to solve it quickly by maintaining coordination between them.

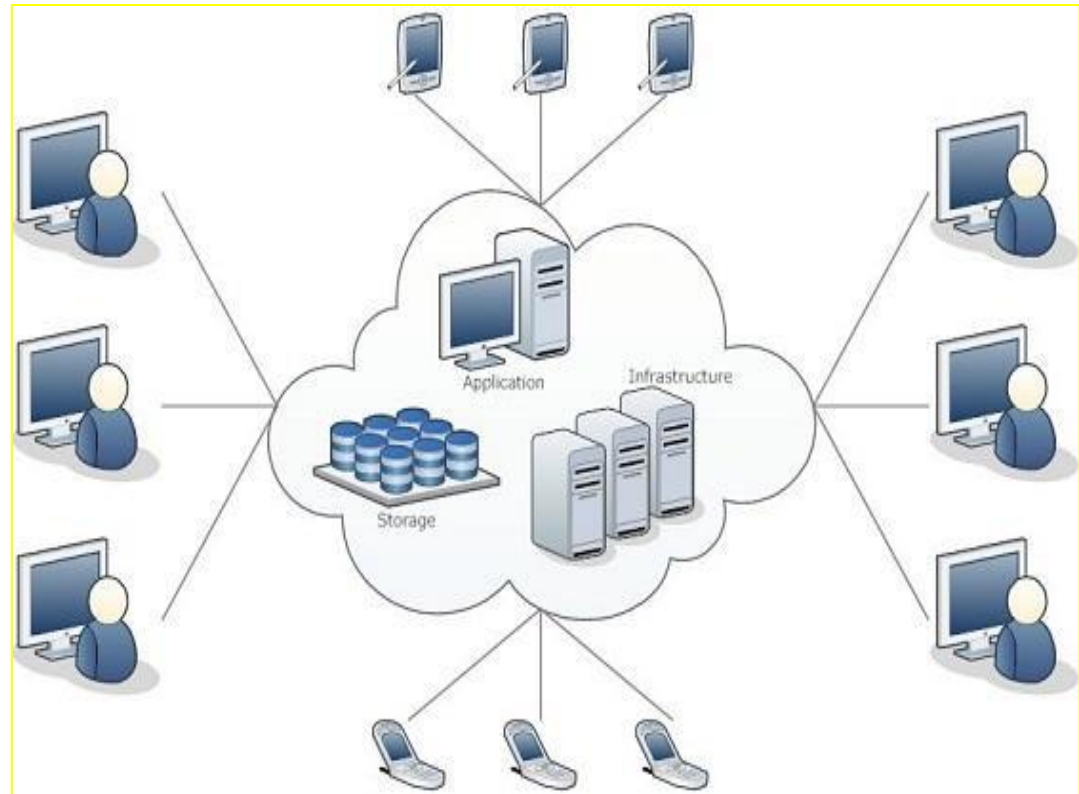
07.	Some disadvantage of cloud computing includes less control especially in the case of public clouds , restrictions on available services may be faced and cloud security.	Some disadvantage of cloud computing includes chances of failure of nodes , slow network may create problem in communication.
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Basic Concepts

There are certain services and models working behind the scene making the cloud computing feasible and accessible to end users.

Following are the working models for cloud computing:

Deployment Models
Service Models



Deployment models define **how the cloud is accessed and where it is located**. There are four main types:

1. Public Cloud

- Accessible to the general public.
- Cost-effective and scalable.
- Less secure due to openness.

Example: Email services like Gmail, Outlook.

2. Private Cloud

- Accessible only within a single organization.
- Provides higher security and control.
- Typically more expensive than public cloud.

Example: A bank's internal cloud for financial data.

3. Community Cloud

- Shared by a group of organizations with similar requirements.
- Cost and resources are distributed among the community.

Example: Government organizations sharing a common cloud for public services.

4. Hybrid Cloud

Combination of public and private clouds.

Critical activities are handled by the private cloud.

Non-critical activities are handled by the public cloud.

Example: An e-commerce company using private cloud for payment processing and public cloud for customer interactions.

SERVICE MODELS

Service models form the **fundamental reference framework** on which cloud computing is built. They are primarily categorized into three core models:

- **Infrastructure as a Service (IaaS)**
 - **Platform as a Service (PaaS)**
 - **Software as a Service (SaaS)**

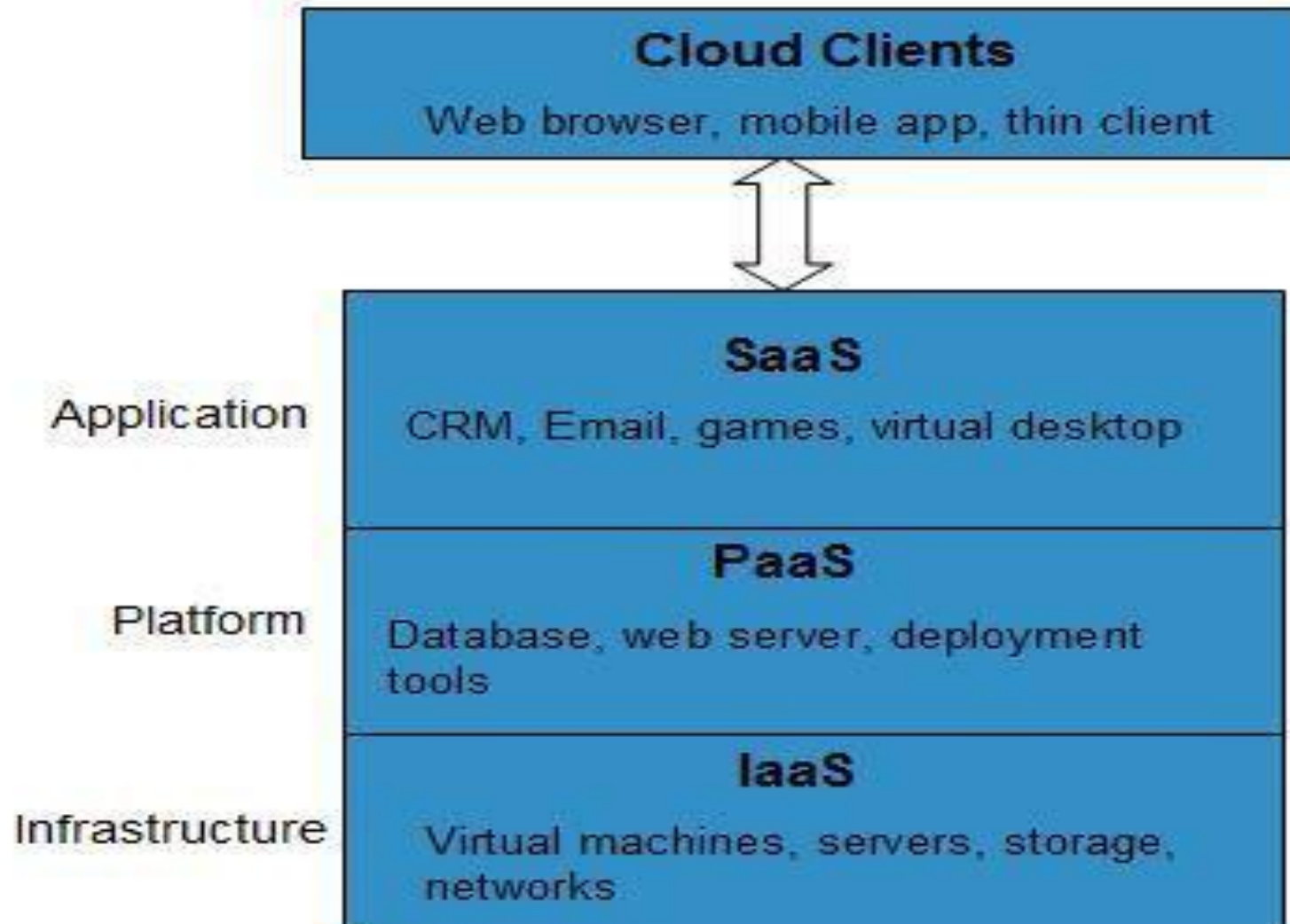
In addition to these, several other service models have emerged under the umbrella term **XaaS (Anything as a Service)**.

Examples include

- **Network as a Service (NaaS),**
- **Business as a Service (BaaS),**
- **Identity as a Service (IDaaS),**
- **Database as a Service (DBaaS),**
- **Strategy as a Service (STaaS).**

Among these, **Infrastructure as a Service (IaaS)** represents the most fundamental layer, providing the essential computing, storage, and networking resources. Each higher-level model is built on top of the lower one, thereby **inheriting the security, scalability, and management mechanisms** of the underlying service model.

SERVICE MODELS

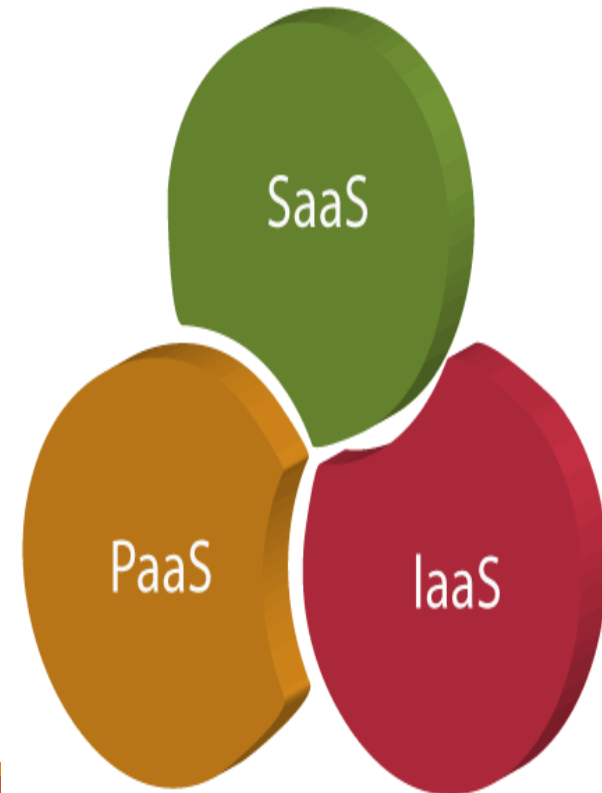


SERVICE MODELS

- INFRASTRUCTURE AS A SERVICE (IAAS)
IaaS provides access to fundamental resources such as physical machines, virtual machines, virtual storage, etc.
- PLATFORM AS A SERVICE (PAAS)
PaaS provides the runtime environment for applications, development & deployment tools, etc.
- SOFTWARE AS A SERVICE (SAAS)
SaaS model allows to use software applications as a service to end users.

SERVICE MODELS

- Infrastructure as a Service (IaaS)
- Platform as a Service (PaaS)
- Software as a Service (SaaS)



INFRASTRUCTURE AS A SERVICE

IaaS is also known as **Hardware as a Service (HaaS)**. It is a computing infrastructure managed over the internet. The main advantage of using IaaS is that it helps users to avoid the cost and complexity of purchasing and managing the physical servers.

Key features:

Instead of purchasing hardware outright, users pay for IaaS on demand. Infrastructure is scalable depending on processing and storage needs. Saves enterprises the costs of buying and maintaining their own hardware.

Because data is on the cloud, there can be no single point of failure.

Enables the virtualization of administrative tasks, freeing up time for other work.

Example: Amazon Web Services (AWS), Microsoft Azure, Google Compute Engine (GCE), Rackspace, Cisco Metacloud.

PLATFORM AS A SERVICE

PaaS cloud computing platform is created for the programmer to develop, test, run, and manage the applications.

Key features:

PaaS provides a platform with tools to test, develop, and host applications in the same environment.

Enables organizations to focus on development without having to worry about underlying infrastructure.

Providers manage security, operating systems, server software and backups.

Facilitates collaborative work even if teams work remotely.

Example: Windows Azure, Google App Engine, Apache Stratos.

SOFTWARE AS A SERVICE

SaaS is also known as "**on-demand software**". It is a software in which the applications are hosted by a cloud service provider. Users can access these applications with the help of internet connection and web browser.

Key features:

SaaS vendors provide users with software and applications via a subscription model.

Users do not have to manage, install or upgrade software; SaaS providers manage this.

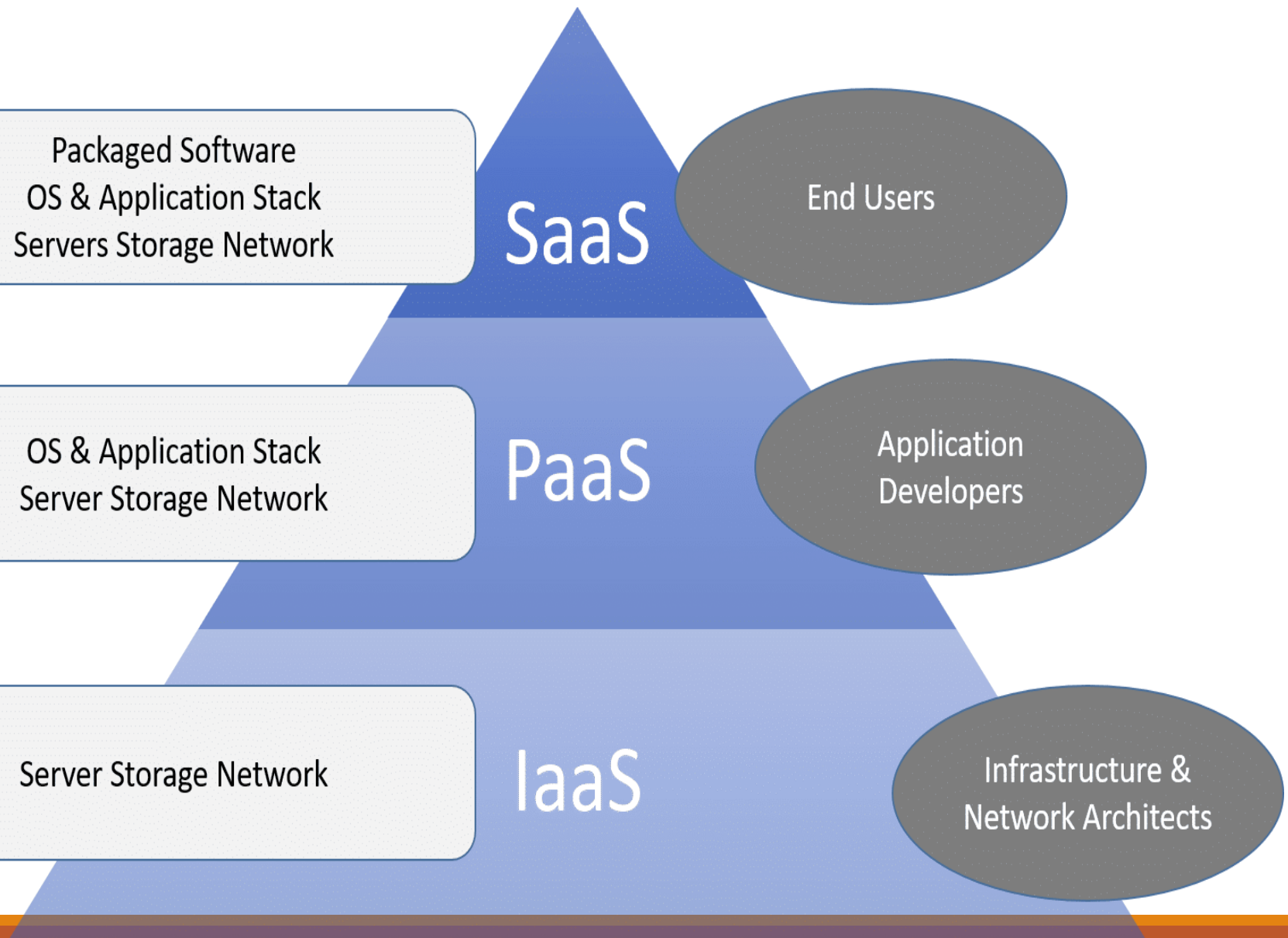
Data is secure in the cloud; equipment failure does not result in loss of data.

Use of resources can be scaled depending on service needs.

Applications are accessible from almost any internet-connected device, from virtually anywhere in the world.

Example: BGoogle Apps, Salesforce, Dropbox, ZenDesk, Cisco WebEx.

Cloud Service Models



RISKS

1. Security & Privacy

- Biggest concern in cloud computing.
- Data and infrastructure are managed by third-party providers.
- Storing sensitive information externally increases risk.
- Any breach could lead to loss of client trust and business.

Example: A hacker gaining access to customer financial data stored in the cloud.

2. Vendor Lock-in

- Difficult to switch from one Cloud Service Provider (CSP) to another.
- Creates dependency on a single provider.

Example: A company using AWS may find it costly and complex to migrate to Azure or Google Cloud.

3. Isolation Failure

- Cloud providers use **multi-tenancy** (resources shared by many users).
- Failure in separation of storage, memory, or routing could expose data to other tenants.

Example: A bug in virtualization software could let one customer access another's data.

4. Management Interface Compromise

- In public cloud, management interfaces are Internet-accessible.
- If compromised, attackers can control cloud services.

Example: A weakly secured admin panel being hacked.

5. Insecure or Incomplete Data Deletion

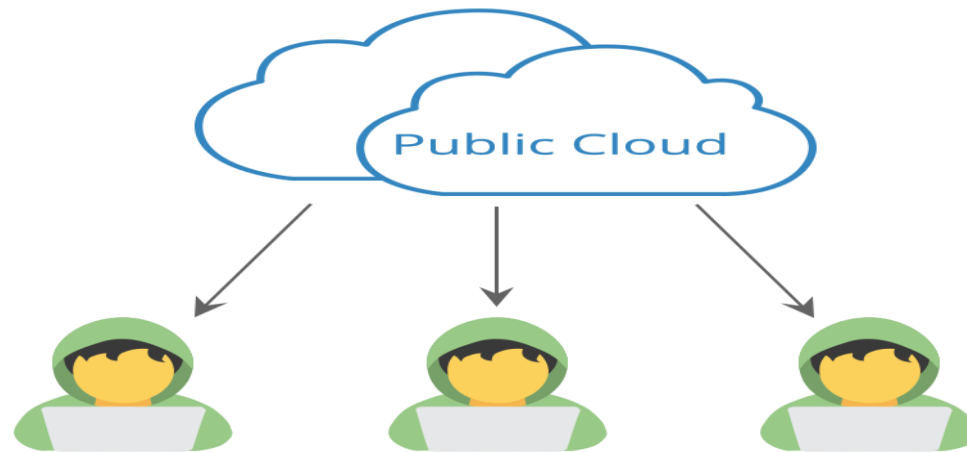
- Data requested for deletion may not be fully removed.
- Extra backup copies or shared disks may still hold fragments of data.

Example: A company deletes customer records, but backups stored in multiple locations still contain them.

PUBLIC CLOUD DEPLOYMENT

- **Public Cloud** is a cloud where services are provided by **third-party providers** over the internet for public use.
- **Accessibility:** Available to the **general public**—anyone can access it.
- **Data Storage:** Data is created, managed, and stored on the **provider's servers**, not on the user's own servers.
- **Ownership:** The **cloud service provider** owns and operates all the hardware, software, and supporting infrastructure.
- **Use Cases:** Commonly used for **applications, file sharing, email services, testing environments, and storing non-sensitive data**.
- **Examples:** Google Drive, Microsoft OneDrive, Dropbox, Amazon Web Services (AWS).

PUBLIC CLOUD



- They keep devices in massive data centers.
- This model also plays a vital role in development and testing.
- That's because it's cheap, easily configurable, can be deployed quickly, and supports fluctuating demands

Advantages

Low Cost

- No need to buy expensive **hardware or software**—the service provider covers the infrastructure.
- Works on a “**pay-per-use**” **model**, so you only pay for what you use, avoiding extra costs.

Convenient Infrastructure Management

- No need to worry about **installation, upgrades, or maintenance**—the provider manages everything.
- Setup is simple, and services are ready to use quickly.

Time-Saving

- Faster **development, testing, and deployment** of applications.
- Provider’s **global network of servers** ensures 24×7 availability and reliability.

Disadvantages

- Data and Security Risks – Being a public platform, data is stored on shared resources, which makes it more vulnerable to threats. Users often have little control or visibility over where their data is stored and who can access it.
- Network Performance Issues – With a large number of users accessing the cloud simultaneously, performance can fluctuate, leading to instability and slower response times.
- Limited Customization – Since the infrastructure is designed for the general public, options for personalization and customization are restricted compared to private or hybrid models.

Private Cloud Deployment

Private Cloud



- A **Private Cloud deployment model**, also known as an *internal* or *corporate cloud*, is dedicated to a single organization.
- In this model, the organization has full control over the systems and their management. While third-party providers can host a private cloud, many companies prefer to maintain their own on-premises data centers, where an in-house IT team manages and oversees everything.
- The Private Cloud is the **second most widely used deployment model** and comes with its own set of advantages and disadvantages.
- A **Private Cloud** is a type of cloud computing environment that is **dedicated exclusively to a single organization**. Unlike public clouds, where resources are shared among multiple users, a private cloud provides **exclusive access, enhanced control, and greater security**.

PRIVATE CLOUD ADVANTAGES

Advantages of Private Cloud

Customization – Unlike public clouds, private clouds allow organizations to tailor solutions according to their specific business needs.

Data Security – Since only authorized internal personnel have access, data security is much stronger. It's an ideal option for storing sensitive corporate information. Additionally, resources can be isolated within the same infrastructure for enhanced protection.

Full Control – The organization retains complete ownership and control over IT operations, service integrations, policies, and user practices.

Support for Legacy Systems – Private clouds can support older applications that may not be compatible with public cloud environments.

Disadvantages of Private Cloud

- **High Cost** – Organizations must invest heavily in hardware and software. On top of that, costs for in-house staff, training, and ongoing maintenance add up, making it a more expensive option compared to public cloud.
- **Limited Scalability** – The scalability of private clouds is restricted by the organization's internal infrastructure and the capacity of the underlying hardware. This means expansion is slower and more resource-dependent than in public cloud environments.

HYBRID CLOUD

- A **hybrid cloud** is a deployment model that combines two or more cloud environments—such as public, private, or community clouds—into a single, unified architecture.
- This model allows organizations to flexibly move data and applications across different clouds based on their needs.
- For example, critical or sensitive operations can be managed in a private cloud, while less sensitive tasks or large-scale workloads can be handled in a public cloud.



HYBRID CLOUD ADVANTAGES

Cost-Effectiveness – By leveraging the public cloud for most data storage and operations, organizations reduce overall costs while only using private cloud resources for critical tasks.

Enhanced Security – Data can be segmented and stored in secure private environments, reducing the risk of theft or unauthorized access.

Flexibility – Hybrid clouds provide the ability to design customized solutions tailored to business requirements, allowing smooth scaling between public and private environments.

COMMUNITY CLOUD

- A **community cloud** is similar to a private cloud, but instead of being dedicated to a single organization, it is shared by multiple organizations with common goals, interests, or requirements.
- In a **private cloud**, one company owns and manages the infrastructure.
- In a **community cloud**, multiple organizations jointly share and manage the infrastructure and resources.

This model is often used by:

- Joint business organizations
- Research groups
- Government agencies
- Industry-specific groups (e.g., healthcare, finance, education)



COMMUNITY CLOUD ADVANTAGES

- **Cost** – A community cloud is a cheaper way to avail the benefits of a private cloud. That's because multiple companies can share the bill together.
- ▣ • **Data Sharing and Collaboration** – With a community cloud, sharing data becomes easy and its collaborative space allows clients to enhance their efficiency. Aside from that, configuration and protocols within a community system meet the needs of the specific industry.
- ▣ **Disadvantages**
 - ▣ • **Limited Storage and Bandwidth**– Within community systems, limited storage and bandwidth capacity become a common problem.

Characteristics of Private Cloud

- **Standardized Solution** – Provides a consistent and uniform cloud environment within the organization.
- **High Virtualization** – Offers the highest degree of virtualization, allowing efficient use of resources.
- **Restricted Data Access** – Data can be restricted and accessed only by a defined group or department.
- **Service Quality Control** – Organizations can define and manage the quality of end services.
- **Need-Based Pricing** – Costs are calculated based on actual usage, making it more transparent.
- **High Customizability** – Solutions can be tailored to meet specific organizational requirements.
- **Enhanced Internal Security** – Provides strong security since resources and access are managed internally.

Building Blocks

Cloud Service Management is like the **control center** of the private cloud.

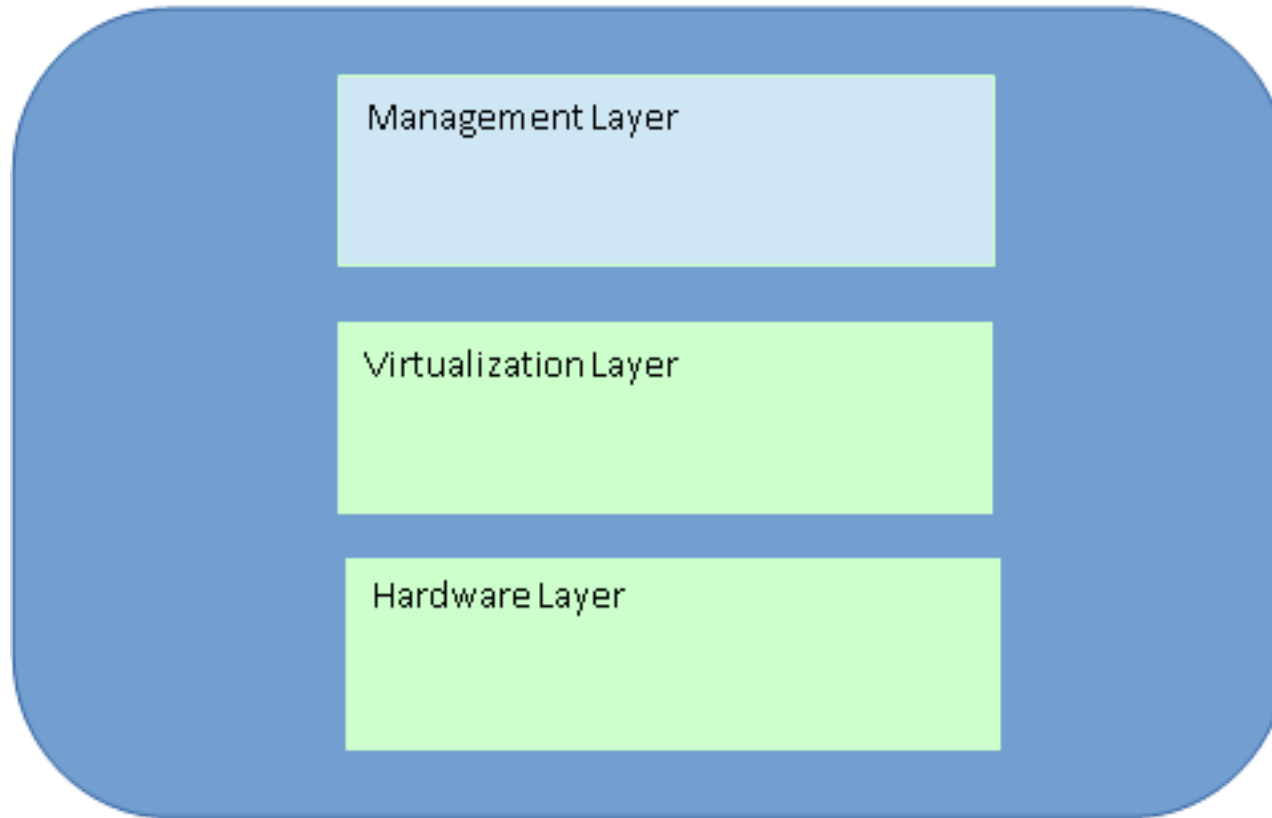
It helps organizations **manage, monitor, and deliver cloud services** to users smoothly.

It ensures that every user or department gets the resources (like storage or virtual machines) they need **on-demand**, without waiting for IT support.

It also follows company rules and policies (for example, how much storage each user can use or who can access certain data).

This layer makes sure that cloud services are provided **securely, quickly, and efficiently**.

Building Blocks of Private Cloud



Infrastructure Automation (Physical or Hardware Layer)

This is the **foundation** of the private cloud.

It includes **real, physical machines** like servers, storage drives, and network systems that run in a data center.

These machines need to work automatically and connect smoothly with the higher layers (like virtualization and management).

Automation tools are used to manage power, cooling, performance, and maintenance — so everything runs efficiently and reliably.

Companies like **HP** and **Dell** provide strong, ready-made hardware systems for this purpose.

Virtualization Layer

The **virtualization layer** plays a key role in achieving essential cloud characteristics such as **resource pooling** and efficient utilization of infrastructure.

Benefits of Virtualization Layer

On-Demand Service Provisioning / Self-Service

A core feature of private cloud environments.

Users can **deploy new services independently** or with minimal support from IT administrators.

Multi-Tenancy

Enables the cloud to **securely support multiple computing environments** on the same infrastructure.

Ensures data isolation and security between tenants.

Simple Resource Scaling

As business needs grow, additional resources (e.g., **RAM, storage**) can be easily added.

Provides flexibility to scale resources **up or down** without major system disruptions.

The **Cloud Management Layer (CML)** is responsible for managing, monitoring, and optimizing cloud resources and services. It sits above the virtualization and infrastructure layers and ensures that cloud operations run smoothly.

Resource Management

- Allocates, monitors, and optimizes cloud resources (CPU, memory, storage, and network).
- Prevents underutilization or overutilization of resources.

Service Provisioning

- Automates the deployment of cloud services.
- Provides **self-service portals** so users can request services without IT intervention.

Monitoring & Reporting

- Tracks system performance, service usage, and resource consumption.
- Generates reports for capacity planning, auditing, and billing.

Security & Compliance

- Implements **identity and access management (IAM)**.
- Enforces compliance with policies, standards, and regulatory requirements.

Billing & Cost Management

- Enables **pay-per-use** or subscription-based pricing models.
- Provides cost transparency for organizations to monitor spending.

Automation & Orchestration

- Automates repetitive IT tasks such as scaling, backup, and recovery.
- Orchestrates workflows across multiple cloud services.

When to Opt for **Public Cloud**

Cost is a Priority

If you don't want to invest heavily in hardware/software.

Best for startups, small businesses, or short-term projects.

Workload is Variable or Temporary

Ideal when your resource needs fluctuate (e.g., seasonal demand, event-based apps).

You can scale up/down quickly and pay only for what you use.

Fast Deployment is Needed

When you want to launch apps quickly without worrying about infrastructure setup.

Collaboration is Important

If many people (employees, clients, or the public) need to access apps or files easily from anywhere.

Non-Sensitive Data Storage

Best suited for storing data that doesn't require strict security or compliance (e.g., file sharing, test environments, public apps).

Private Cloud Deployment Models

A **private cloud** is dedicated to a single organization. It can be managed internally or by a third party. Deployment models here are about **where and how the private cloud is hosted**:

1. On-Premises Private Cloud

1. Cloud infrastructure is hosted **within the organization's own data center**.
2. Example: A bank sets up its own private cloud to securely store customer data.

2. Externally Hosted Private Cloud

1. Infrastructure is dedicated to one organization but **hosted by a third-party provider** in their data center.
2. Example: A healthcare company rents a private cloud from IBM Cloud but it is not shared with other clients.

3. Managed Private Cloud

1. A **third-party vendor manages and maintains** the private cloud (security, updates, scaling), while it remains dedicated to one organization.
2. Example: VMware Cloud Foundation provides managed private cloud solutions.

4. Virtual Private Cloud (VPC)

1. A logically isolated section inside a **public cloud**, dedicated to a single organization.
2. Example: Amazon VPC (Virtual Private Cloud) lets a company create isolated networks on AWS.

Public Cloud Service Models

A **public cloud** is shared among multiple organizations but managed by cloud providers (like AWS, Azure, GCP). The services fall into **three main models**:

1.IaaS (Infrastructure as a Service)

1. Provides virtualized computing resources: servers, storage, networking.
2. Example: **AWS EC2, Google Compute Engine, Azure Virtual Machines.**

2.PaaS (Platform as a Service)

1. Provides a platform for developers to build, run, and deploy applications without managing infrastructure.
2. Example: **Google App Engine, Heroku, Azure App Service.**

3.SaaS (Software as a Service)

1. Provides ready-to-use applications over the internet.
2. Example: **Gmail, Salesforce, Microsoft 365, Zoom, Netflix.**